

# Arsalan Ghogari

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CS and Business at USC. I build and ship real systems: the software stack for an autonomous submarine, a RAG assistant for 20,000 students, and a voice agent that negotiates over the phone. I keep score on all of it: **0.40 to 0.90 mAP**, **6.2 s to 0.5 s**, **83% to 95%**, **12 N to 30 N**.

## Education

**University of Southern California:** Viterbi Engineering & Marshall Business

**Expected May 2027**

Bachelor of Science, Computer Science/Business Administration; Minor, AI Applications | Study Abroad: University of Bristol, UK (Spring 2026)

**GPA: 3.9** | **SAT: 1550** (Math 800, EBRW 750) | Presidential Scholar, Viterbi Scholar, Dean's List, W.V.T. Rusch Honors, Thematic Option Honors

## Experience

**Software Lead, USC RoboSub (Autonomous Underwater Vehicle Team)**

**Jan 2025 - Present**

- **Promoted 3x in 18 months** (Engineer → Perception Lead → Perception & Autonomy Lead → Software Lead); now direct a **5–10 engineer team** owning the AUV's full autonomy stack: vision, SLAM, planning, and controls (ROS 2, NVIDIA Jetson).
- More than doubled underwater detection accuracy (**0.40 → 0.90 mAP@50**) by building the dataset and training YOLO and RT-DETR, then compiling ONNX → TensorRT (FP16/INT8) for real-time on-vehicle inference; separately migrated the team's inherited **10,000+ line ROS 1 codebase** to ROS 2.
- Mined **75 GB of test data across 9 in-water sessions**: found a thrust clamp misconfigured at **12 N instead of 30 N**, recovering roughly **40% of top speed**, and proved the drag model was only **13% off** rather than broken, pulling an unneeded recalibration off the pre-competition critical path.
- Delivered SLAM, fusing stereo (ZED), IMU, DVL, and depth in a GTSAM factor graph and validating it to **under 0.3 m absolute error**; built NVIDIA Isaac Sim environments so the full stack regression-tests closed-loop before scarce in-water time.

**Director of Digital Strategy, USC Undergraduate Student Government**

**May 2025 - Present**

- Own the digital platform serving USC's **20,000+ undergraduates** and lead a 2–3 person team; **re-appointed for a second year** in recognition of technical contributions and impact.
- Made **\$352K in student-government spending** publicly searchable with automated Funding, Senate Bill, and Legislative Project trackers, cutting manual reporting **~95%** across 40+ organizations, committees, and assemblies.

**Suicide Prevention Volunteer & Outreach, Didi Hirsch Mental Health Services**

**Jun 2025 - Present**

- Completed **50+ hours** of intensive crisis-intervention training and answered **100+ crisis calls** on a weekly shift; now on the outreach team expanding community awareness of crisis resources.

## Technical Projects

**Ask USG: Production RAG Chatbot for USC Student Government** | Node/Express, OpenAI, Postgres/pgvector, LangGraph, Docker

- Raised answer quality **83% → 95%** across **1,100+ beta questions** from **80+ USG officers and students**, lifting retrieval **MRR 0.91 → 0.98** with a self-built eval harness; cut time-to-first-token **92% (6.2 s → 0.5 s)**. Ingests all **97** site pages weekly via GitHub Actions; launching campus-wide to **20,000+ students** in Fall 2026.
- Wrote a **30-test suite** proving the mental-health crisis pathway fires across **16 phrasings** and routes to the 988 Lifeline with zero knowledge-base leakage.

**Parley: AI Voice Negotiation Agent (solo build)** | ElevenLabs × Hack-Nation Global AI Hackathon

- A voice agent that finds vendors via live Google Places, phones them, extracts itemized quotes into strict JSON, and negotiates using competing bids as leverage. In a live call the vendor confirmed a mid-negotiation drop from **\$2,400 to \$2,250**; a per-turn tripwire validates every spoken price against stored quotes, backed by **41 CI-gated tests**.

**Collaborative Note-Taking Application** | Java, XML, Android Studio

- Selected as a **top-2 implementation out of 37 teams**; delivered an app **70% faster** than peers via Android Profiler-guided optimization, thread offloading, and database indexing.

## Skills

**Programming:** Python, C++, CUDA, Java, JavaScript, TypeScript, C, MATLAB, HTML/CSS

**AI/ML & Robotics:** PyTorch (YOLO, RT-DETR), TensorRT, ONNX, ROS 2, GTSAM, NVIDIA Isaac Sim, stereo vision (ZED), Foxglove, OpenAI API, LangGraph, Langfuse, RAG (pgvector)

**Web & Infrastructure:** React, Next.js, Node/Express, PostgreSQL, Supabase, WordPress, Docker, K3s, GitHub Actions CI/CD, Git, Linux/Bash, NVIDIA Jetson

**Interests:** weightlifting, weekend group cycling along the Santa Monica coast, underwater robotics, AI hackathons, mental health advocacy